

CANDIDATE
NAME

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PDG

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PDG
INDEX NUMBER

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BIOLOGY 9648/02

Paper 2 Core Paper

13 September 2016
Tuesday

2 hours

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your name and PD group on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graph or rough working.
Do not use paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions.

Section B

Answer any **one** question.

All working for numerical answers must be shown.
At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

Calculators may be used

For Examiner's Use	
Section A	80
1	
2	
3	
4	
5	
6	
7	
8	
Section B	20
9 / 10	
Total	100

Section A

Answer **all** the questions in this section.

- 1** The enzyme human immunodeficiency virus (HIV) protease is essential for the life cycle of HIV.

Fig. 1.1 shows the structure of the HIV protease. It exists as a homodimer made up of two identical polypeptide chains, each forming a subunit. The active site lies between the subunits, with each subunit contributing an aspartic acid that functions as a catalytic residue.

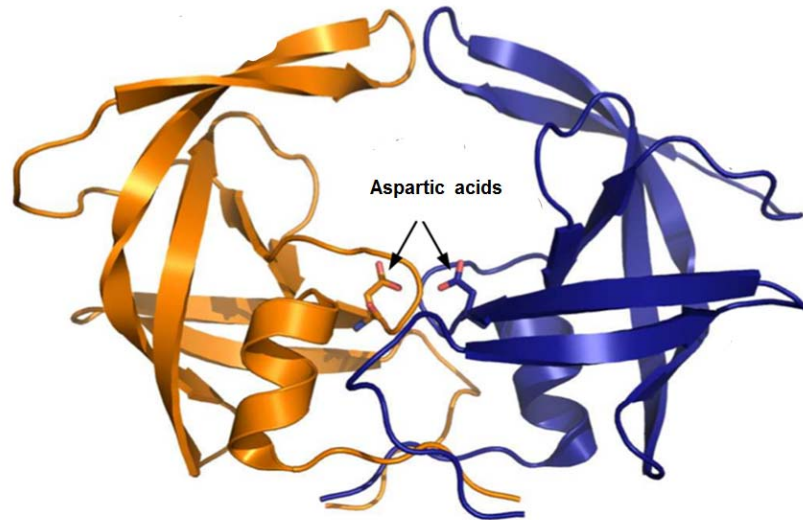


Fig. 1.1

- (a)** (i) Describe the function of aspartic acid in HIV protease.

..... [1]

- (ii) With reference to Fig. 1.1, outline how the structure of HIV protease is formed.

.....

 [4]

- (iii) Suggest an advantage of protease being a homodimer.

.....
 [1]

- (b) Many HIV protease inhibitors have been developed to treat people infected with HIV. Most inhibitors act as competitive inhibitors to HIV protease. Fig. 1.2 shows three such inhibitors.

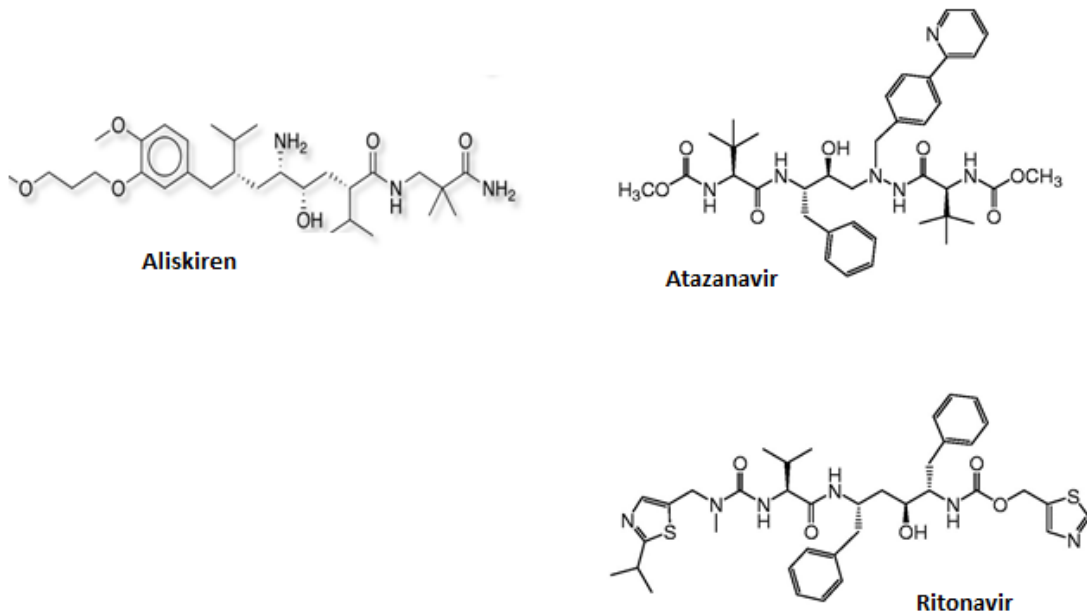


Fig. 1.2

With reference to Fig. 1.2, explain how HIV protease inhibitors work in the treatment of HIV infection.

.....

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.....

.....

..... [3]

[Total: 9]

- 2 Fig. 2.1 shows an electron micrograph of an actively dividing cell from *Bellevalia romana*, a flowering herbaceous plant.

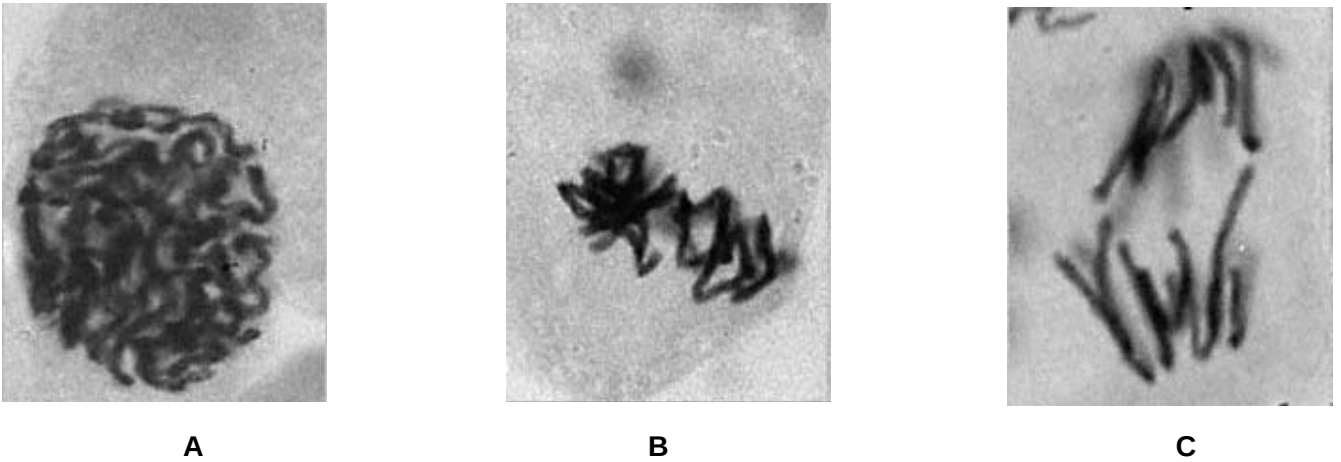


Fig. 2.1

- (a) (i) Identify stages **A** and **B**, and state the visible features that enabled your identification.

.....

 [2]

- (ii) Explain the change in distance between centromeres of a chromosome in stage **C**.

.....

 [3]

- (iii) Explain why it is important that replication occurs before mitosis.

.....

 [2]

- (iv) Explain how homologous chromosomes in stage **A** are genetically different from those in prophase II of the same plant.

.....

.....

.....

.....

..... [3]

Prokaryotic organisms such as *Escherichia coli* do not divide by mitosis. Apart from ribosomes, prokaryotes have no organelles comparable to those found in eukaryotes and have a circular 'chromosome' with no centromere.

- (b) With reference to the information given above and your knowledge of mitosis, suggest why mitosis does not occur in prokaryotes.

.....

.....

..... [2]

[Total: 12]

3 Fig. 3.1 is a diagram showing translation. Table 3.2 shows an mRNA codon table.

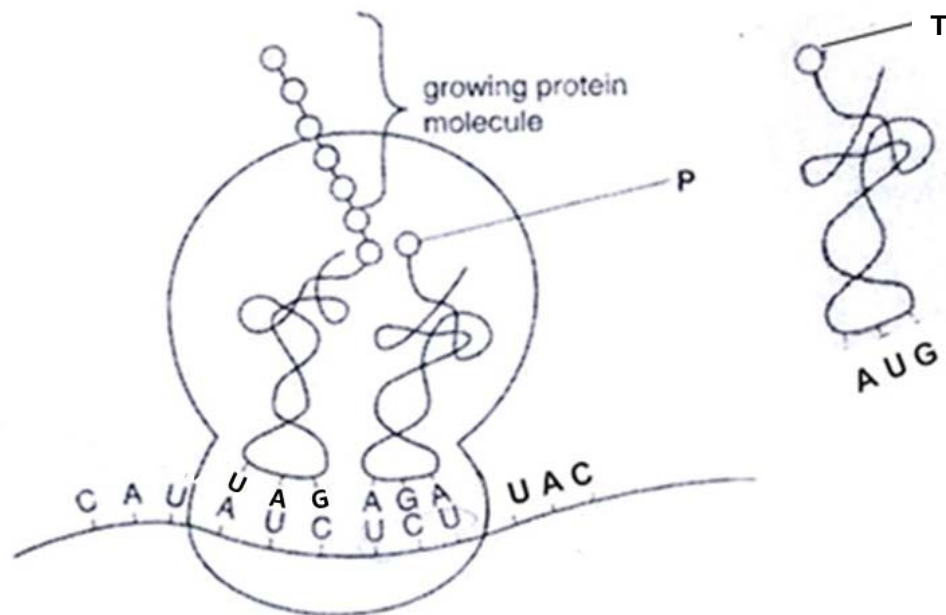


Fig. 3.1

		Second Base				
		U	C	A	G	
First Base	U	phenylalanine	serine	tyrosine	cysteine	U
		phenylalanine	serine	tyrosine	cysteine	C
		leucine	serine	(stop)	(stop)	A
		leucine	serine	(stop)	tryptophan	G
	C	leucine	proline	histidine	arginine	U
		leucine	proline	histidine	arginine	C
		leucine	proline	glutamine	arginine	A
		leucine	proline	glutamine	arginine	G
	A	isoleucine	threonine	asparagine	serine	U
		isoleucine	threonine	asparagine	serine	C
		isoleucine	threonine	lysine	arginine	A
		(start) methionine	threonine	lysine	arginine	G
	G	valine	alanine	aspartate	glycine	U
		valine	alanine	aspartate	glycine	C
		valine	alanine	glutamate	glycine	A
		valine	alanine	glutamate	glycine	G
	G	valine	alanine	aspartate	glycine	U
		valine	alanine	aspartate	glycine	C
		valine	alanine	glutamate	glycine	A
		valine	alanine	glutamate	glycine	G

Table 3.2

- (a) (i) State **two** molecules required for translation that are not shown in Fig. 3.1.

..... [1]

- (ii) Using Table 3.2 to identify **P** and **T**, briefly describe the process of translation as shown in Fig. 3.1.

.....
.....
.....
.....
.....
..... [4]

- (b) Suggest why parts of tRNA are double stranded.

.....
.....
..... [2]

- (c) State **two** ways in which DNA replication:

- (i) is similar to transcription.

.....
.....
..... [2]

- (ii) differs from transcription.

.....
.....
..... [2]

[Total: 11]

4

(a) Explain why viruses are obligate parasites.

.....

.....

..... [2]

Fig. 4.1 is an electron micrograph of T4 bacteriophages infecting an *Escherichia coli*.

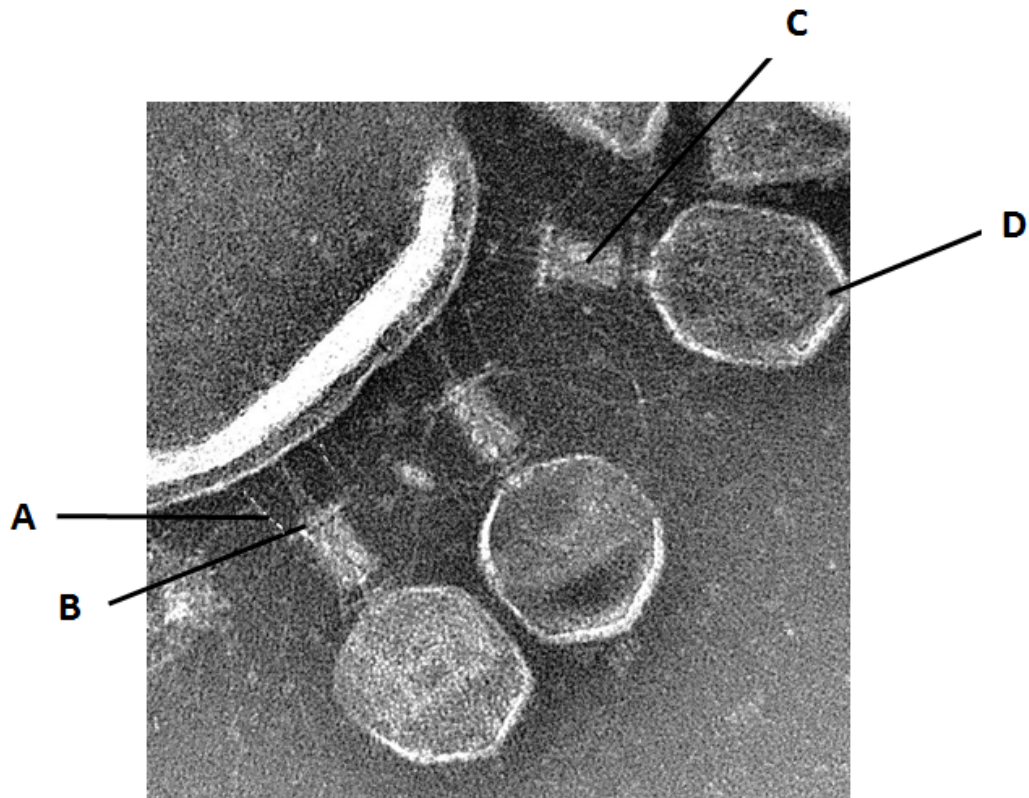


Fig. 4.1

(b) (i) Identify the structures labelled A - D.

A.....

B.....

C.....

D..... [2]

- (ii) Explain why polymerase enzymes are present in human influenza virus but not in T4 bacteriophage.

.....

.....

..... [2]

- (c) With reference to the reproductive cycle of bacteriophages, suggest why bacteriophage infection may be beneficial to bacteria population.

.....

.....

.....

.....

..... [3]

[Total: 9]

- 5 Voltage-gated sodium ion channels are integral membrane proteins that allow generation of an action potential across the axon membrane.

Fig. 5.1 shows a typical voltage-gated sodium ion channel which has two gates, the activation and inactivation gates. At different membrane potentials, the channel exists in different states depending on which gates are open or close.

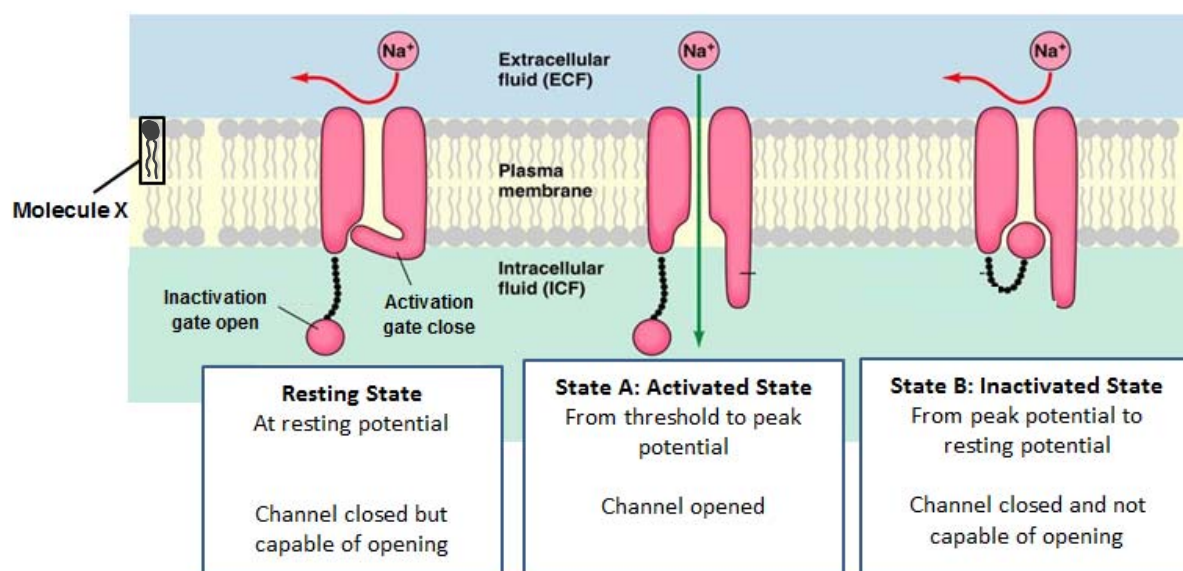


Fig. 5.1

- (a) (i) Describe how State **A** results in depolarisation.

.....

.....

..... [2]

- (ii) Explain how State **B** ensures unidirectional movement of nerve impulse along the neurone.

.....

.....

.....

.....

..... [3]

(iii) Describe how Molecule **X** differs from a triglyceride in terms of structure.

.....

.....

..... [2]

(b) Explain why triglycerides are better respiratory substrates than carbohydrates.

.....

.....

..... [2]

[Total: 9]

- 6** In *Drosophila*, the characteristics of wing length and body colour are controlled by one gene each. Wild type flies have normal (long) wings and grey body colour while mutant flies have vestigial (undeveloped) wings and black body colour. Pure-breeding wild-type and mutant flies will breed to produce flies with normal wings and grey colour.

A male fly with wild phenotype was crossed with a female fly with mutant phenotype. The resulting offspring were as follows:

normal wings and black body colour 107

vestigial wings and grey body colour 92

Using the following symbols:

N normal wings **n** vestigial wings

E grey body colour **e** black body colour

- (a)** Draw a genetic diagram in the space below showing the cross described.

(b) The observed numbers are usually different from the expected numbers in any genetic cross.

- (i)** Suggest **two** reasons why such a difference may occur in a monohybrid cross, referring only to events after meiosis.

.....
.....
..... [2]

- (ii)** A chi-squared test was carried out on the results of the cross. A p value of about 0.30 is obtained.

With reference to the cross results, explain the significance of the p value.

.....
.....
..... [2]

(c) The wing length of fruit flies with normal wings varies between 70 to 85 μm .

Suggest a reason for the variation in wing length within fruit flies with normal wings.

.....
..... [1]

(d) Alleles coding for mutant phenotypes in fruit flies are caused by artificially induced mutations in laboratory bred flies. Suggest why such mutants are unlikely to be found in natural populations.

.....
.....
..... [2]

[Total:11]

- 7 Giant anteaters, armadillos and Australian numbats (*Myrmecobius fasciatus*) have many similar traits. This led some to believe that they were closely related and that they should be classified into the same taxon of a lower rank under the hierarchical classification system.

Table 7.1 shows the comparison of four characteristics between the three mammals

Mammal	Characteristics			
	Diet	Body	Snout	Tongue
Armadillo	Feed on insects	Covered by bony keratinised plates	Pointy snout	Long tongues
Giant Anteater	Feed on ants and termites	Covered by hair	Elongated narrow snout	Long tongues
Numbats	Feed on termites	Covered by hair	Narrow snout	Long tongues

Table 7.1

DNA sequences of selected genes such as 18s rRNA are subsequently compared between some organisms, including the three mammals, when molecular experimental techniques advanced and the results helped clarified the phylogenetic relationships of the mammals.

Fig. 7.2 shows the simplified phylogenetic tree of the organisms based on nucleotide sequence comparison results.

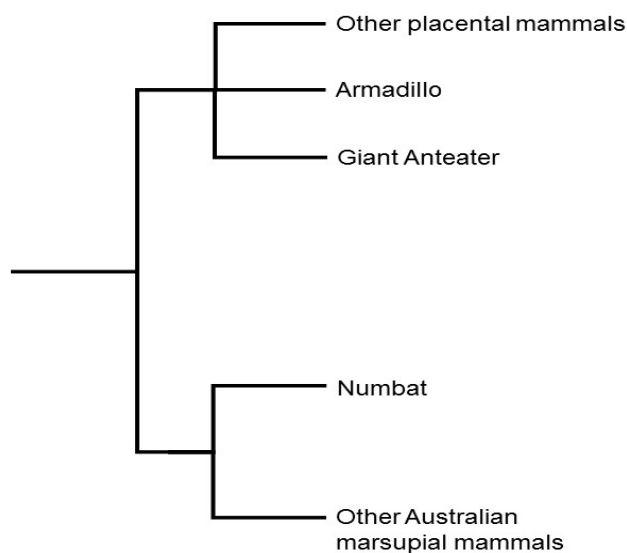


Fig. 7.2

- (a) Explain the relationship between classification and phylogeny.

.....

.....

..... [2]

- (b) Using the information above, explain why comparison of morphological structures led to the incorrect conclusion about the phylogenetic relationships of the three mammals.

.....

.....

.....

.....

.....

..... [4]

- (c) State **one** reason why the 18s rRNA gene was chosen to compare DNA sequences between organisms.

.....

..... [1]

Fig. 7.3 shows the geographical distribution of the various mammals.

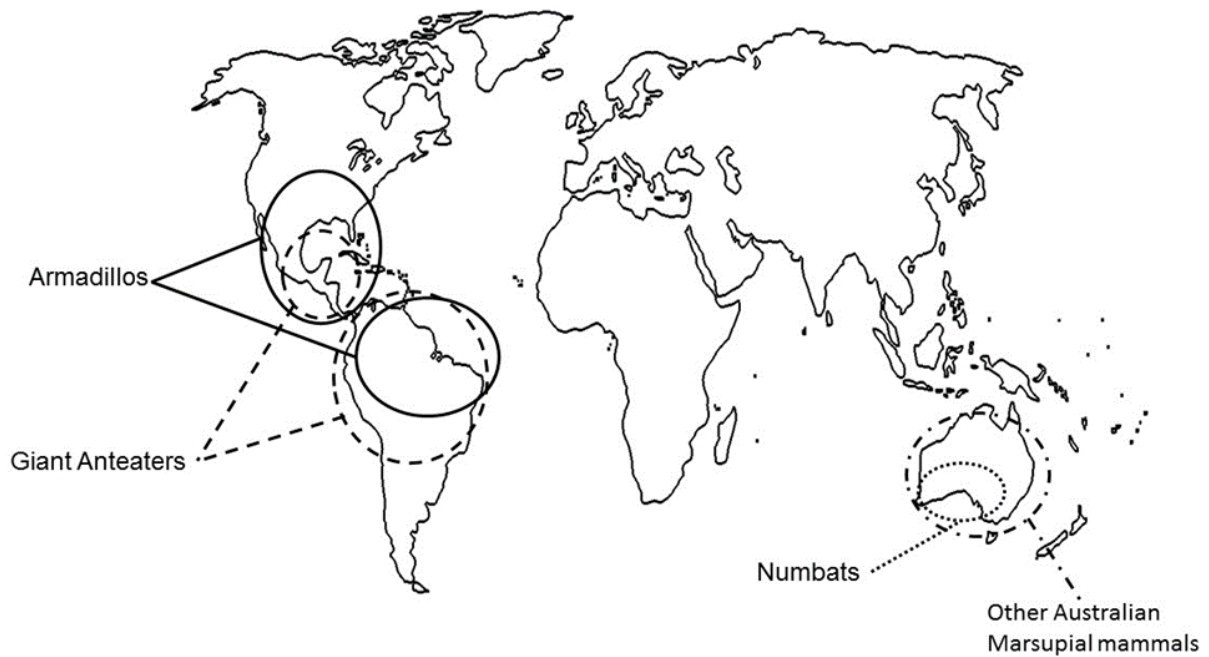


Fig. 7.3

- (d)** Using the information above, explain how biogeography supports the phylogenetic relationships constructed from DNA sequence comparison.

.....

.....

..... [2]

[Total: 9]

- 8 Epidermal growth factor receptors (EGFR), which belong to the ErbB family of receptor tyrosine kinases, have been found to be overexpressed in many cancers, such as glioblastoma, colorectal and breast cancers.

Fig. 8.1 shows the EGFR signalling pathway.

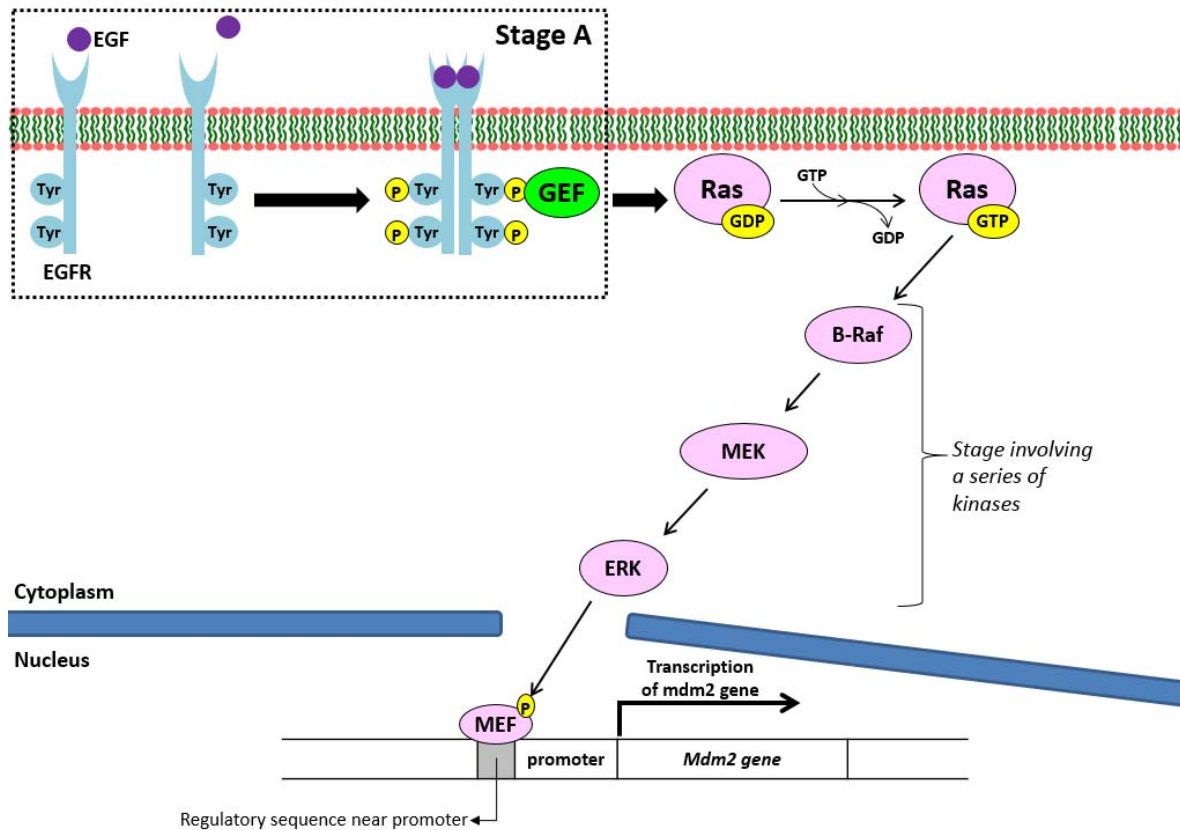


Fig. 8.1

Legend:

EGF	Epidermal growth factor
Ⓟ	Phosphate group
GEF	Guanine nucleotide exchange factor that binds to and activates Ras
B-Raf, MEK, ERK	Kinases
MEF	A specific transcription factor that binds near the promoter

With reference to Fig. 8.1,

- (a)** Describe the events occurring at stage A.

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.....

.....

.....

..... [4]

- (b)** **(i)** Suggest how a mutation in Ras GTPase that causes GTP to be permanently bound results in the overexpression of mdm2.

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..... [3]

- (ii)** Mdm2 is an enzyme which catalyses the addition of ubiquitin to p53. Explain how high levels of mdm2 enzyme may lead to increased chances of cancerous growth.

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..... [3]

[Total: 10]

Section B

Answer **EITHER 9 or 10.**

Write your answers on the separate answer paper provided.

Your answer should be illustrated by large, clearly labeled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in section (a), (b) etc., as indicated in the question.

- 9** **(a)** Explain how fluidity of biological membranes can be maintained and the importance of fluidity to membrane function. [8]
- (b)** Plant cells have a cellulose cell wall outside the cell surface membrane. Explain how the structure of cellulose is related to its function. [7]
- (c)** Describe how photophosphorylation differs from oxidative phosphorylation. [5]

[Total: 20]

Or

- 10** **(a)** Distinguish between gene mutation and chromosome structural mutation. [4]
- (b)** Describe how the most common CFTR gene mutation affects function of the protein and explain why other mutations vary in the extent to which they affect protein function. [8]
- (c)** With reference to mutation of named genes, outline the development of cancer. [8]

[Total:20]